```python
#Determine user input
something = input("Enter something : ")
print("You entered : ", something)
print("Data type of num : ", type(something))
```

```
Enter something : fhsjigvhuhb
You entered :  fhsjigvhuhb
Data type of num :  <class 'str'>
```

```python
num = int(input("Enter a number : "))
print("Square : ",(num ** 2))
print("Cube : ",(num ** 3))
print("Square root : ",(num ** 0.5))
```

```
Enter a number : 4
Square :  16
Cube :  64
Square root :  2.0
```

```python
#arithmentic operation
a=int(input("Enter first no : "))
b=int(input("Enter second no : "))
print("addition is : ",(a+b))
print("substraction is : ",(a-b))
print("multiplication is : ",(a*b))
print("division is :",(a/b))
print("modulus is : ",(a%b))
print("floor division is : ",(a//b))
print("exponent is : ",(a**b))
```

```
Enter first no : 1
Enter second no : 2
addition is :  3
substraction is :  -1
multiplication is :  2
division is : 0.5
modulus is :  1
floor division is :  0
exponent is :  1
```

```python
#comparison operator
a=int(input("Enter first no : "))
b=int(input("Enter second no : "))
print((a>b))#greater then
print((a<b))#less then
print((a==b))#equal to
print((a!=b))#not equal to
print((a>=b))#greater then or equal to
print((a<=b))#less then or equal to
```

```
Enter first no : 2
Enter second no : 3
False
True
False
True
False
True
```

```
#area of right angled triangle
l=float(input("Enter length : "))
h=float(input("Enter breadth : "))
print("Area of the right angled triange : ",(0.5*l*b))
```

```
Enter length : 2
Enter breadth : 3
Area of the right angled triange :  3.0
```

```
# average
a=float(input("Enter first no : "))
b=float(input("Enter second no : "))
c=float(input("Enter third no : "))
avg = (a+b+c)/3
print("Average : ",avg)
```

```
Enter first no : 1
Enter second no : 2
Enter third no : 3
Average :  2.0
```

```
#simple interest
p=float(input("Enter principal value : "))
r=float(input("Enter interest rate : "))
n=float(input("Enter no of year: "))
si = (p*r*n)/100
print("Simple Interest : ",si)
```

```
Enter principal value : 1000
Enter interest rate : 5
Enter no of year: 3
Simple Interest :  150.0
```

```
# celsius to fahrenheit
c = float(input("Enter the temperature in celsius : "))
f = ((9/5)*c)+32
print("Temperature in Fahrenheit : ",f)
```

```
Enter the temperature in celsius : 29
Temperature in Fahrenheit :  84.2
```

```python
# euclidean distance
x1 = float(input("Enter first x co-ordinate : "))
y1 = float(input("Enter first y co-ordinate : "))
x2 = float(input("Enter second x co-ordinate : "))
y2 = float(input("Enter second y co-ordinate : "))
dist = ((((x2-x1)**2)+((y2-y1)**2)) ** 0.5)
print("Euclidean distance : ",dist)
```

```
Enter first x co-ordinate : 1
Enter first y co-ordinate : 2
Enter second x co-ordinate : 3
Enter second y co-ordinate : 4
Euclidean distance :  2.8284271247461903
```

```python
# swap 2 variables using comma,temp,bitwiseXOR
a=3
b=4
a,b=b,a
print(a,b)
```

```
4 3
```

```python
a=3
b=4
t=a
a=b
b=t
print(a,b)
```

```
4 3
```

```python
a=9
b=4
a= a^b
b = a^b
a = a^b
print(a,b)
```

```
4 9
```

```python
# solution to quadratic equation
a=float(input("Enter first co-effiecient : "))
b=float(input("Enter second co-effiecient : "))
c=float(input("Enter third co-effiecient : "))
D = ((b**2)-(4*a*c))
if D > 0:
    x1=(((b*(-1))+(D ** 0.5))/(2*a))
    x2=(((b*(-1))-(D ** 0.5))/(2*a))
    print("Roots of the equation are : ",x1,"and",x2)
elif D==0:
    x=((b*(-1))/(2*a))
```

```
    print("Root of the equation is : ",x)
else:
    print("Real solution doesn't exist")
    x1=(((b*(-1))+(D ** 0.5))/(2*a))
    x2=(((b*(-1))-(D ** 0.5))/(2*a))
    print("Complex Roots of the equation are : ",x1,"and",x2)
```

```
Enter first co-effiecient : 1
Enter second co-effiecient : 1
Enter third co-effiecient : 1
Real solution doesn't exist
Complex Roots of the equation are :  (-
0.49999999999999994+0.8660254037844386j) and (-0.5-
0.8660254037844386j)
```